Orphans and Archetypes: Linking Unidentified Ritual Fragments to Hittite Textual Traditions through Quantitative Scribal Tradition Mapping

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Abstract

The Hittite ritual corpus contains a large body of manuscript fragments that resist classification within established textual traditions: Laroche's residual category CTH 470 and the newly digitised EBo fragments, 3,033 manuscripts in the TLHdig 0.3 release, with a median of 10 (CTH 470) and 4 (EBo) legible words. Philological assignment needs a parallel passage, which these "orphans" lack. We treat tradition assignment as a content-based classification problem. Word tf-idf and orthographic sign n-gram features from TLHdig define a reference space of ten ritual and festival traditions, each delimited by the study that argued for it and mapped back to Laroche's CTH numbers. A held-out manuscript is assigned to its correct tradition in 84% of cases (majority baseline 30%), and a ten-word window in 68%. Separating the layers of a text shows that traditions are identified by what is written (logograms, Hittite vocabulary, foreign-language words), not by how it is written: spelling adds nothing beyond the words, and orthographic rates carry only a weak signal that follows compositions and their copies. On CTH 701, where Salvini and Wegner's subgroups provide partial ground truth, the method recovers 58% of subgroup labels (majority 21%) and does not predict script date. With a calibrated abstention margin, 940 of 2,010 orphans are placed; an open-set test shows that a closed reference must be paired with explicit outside classes before such placements can be read as "this tradition". We interpret the results through Quantitative Scribal Tradition Mapping (QTM), which treats orphans as low-coverage witnesses to incompletely attested archetypes.

Keywords

Hittite, cuneiform, stylometry, fragment attribution, open-set recognition, textual traditions, digital philology

1. Introduction

Emmanuel Laroche's *Catalogue des textes hittites* [1] gave every Hittite composition a number, and with it a place: rituals in one block, festivals in another, texts in Hattian, Luwian and Hurrian in blocks of their own. What could not be placed went into residual categories. CTH 470, "fragments of rituals", is the largest of them, and it has grown with every volume of hand copies since 1971. The Mainz Hethitologie-Portal now keeps Laroche's numbers as a shared address system while moving fragments after joins and adding new texts [2, 3]. The digitisation of the EBo fragments in TLHdig 0.3 [4] adds 1,151 more manuscripts with a median of 4 legible words, most of them filed under CTH 832 (562) and CTH 670 (382), two further residual categories.

The philological route to assigning such a fragment runs through a parallel: a duplicate or a close parallel passage in an identified composition. That route is closed for most orphans by definition. This paper asks whether the content of a fragment, its words and the way they are spelled, can point to its tradition instead, and how far such a pointer can be trusted. It builds on an exploratory study of the Hurrian and Luwian festival texts [5], which mapped about 1,000 manuscripts with tf-idf and t-SNE, examined CTH 701 under the "microscope", and announced as its next step the test of which word groups carry the signal. Here the exploration becomes assignment, with explicit baselines, held-out evaluation and a rule for when not to assign.

Our contributions are:

1. a reference space of ten Hittite ritual and festival traditions, each defined by the study that argued for it and mapped back to Laroche's CTH numbers, with the CTH ranges that no study claims set

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aside as test sets (Section 3);
2. a validation of nearest-centroid tradition assignment against majority and permutation baselines, as a function of fragment size, and on CTH 701, where partial ground truth exists (Sections 5.1–5.4);
3. a decomposition of the signal into layers and word classes that separates what is written (cultural content) from how it is written (scribal habit) (Section 5.3);
4. a placement of the CTH 470 and EBo orphans with a calibrated abstention margin, and an open-set test of whether the method notices a fragment that belongs to no reference tradition (Sections 5.5–5.6);
5. Quantitative Scribal Tradition Mapping (QTM), a framework that reads the results in terms of witnesses and archetypes (Section 6).

All code, the label snapshot and the outputs are public, and every number in this paper is generated from the experiment outputs (Section 9).

2. Background and related work

Traditions and the catalogue. Laroche's catalogue sorts by genre first and by language second, while later scholarship defines a ritual tradition by the origin of its practitioners, its cult and its ritual language. One tradition therefore ends up scattered over several CTH ranges. The Kizzuwatna rituals occupy CTH 471–500, but the ritual of the Kizzuwatnean Old Woman Maštigga is filed with the named practitioners at CTH 404 [6]; the rituals of the AZU priest (CTH 701) sit in the Hurrian block rather than with the Kizzuwatna rituals [7]; and the named-practitioner rituals of CTH 390–469 come from Arzawa, the Lower Land and Kizzuwatna alike [8, 9]. The festival texts have been regrouped by cult geography [5, 10, 11] and, in the Hethitische Festrituale (HFR) project, into edition modules such as the Luwian and the Palaic festivals [12, 13]. Surveys of Hittite religion frame all of these [14, 15].

Scribal habit and its measurement. Hittite scribes copied, excerpted and re-edited ritual texts over two centuries, and their writing habits (logographic writing, plene spelling, sign choice) vary with date, school and individual [16, 17, 18, 19, 20]. Whether a habit marks a tradition, a scribal circle or a date is therefore an empirical question, and one this paper tests rather than assumes. Stylometry offers the tools: frequency-based representations, distance measures and attribution protocols with held-out evaluation [21, 22, 23]. Two of its known problems are central here. Short samples degrade attribution sharply [24], and a closed-set classifier always names one of its candidates, even when the true author is not among them [25]; the latter is the problem of open-set recognition [26].

Computational work on cuneiform. Machine learning on cuneiform has addressed transliteration and translation [27, 28], language and dialect identification [29], the matching of fragments to known compositions by sign sequences [30], and palaeographic style [31]. Fragment matching finds a known text; it does not place a fragment whose parallels are lost. All of these studies are closed-set. To our knowledge, no published study uses content features to assign Hittite fragments to religious traditions, or asks when a fragment fits none of the candidates.

3. Data

Corpus. We use the TLHdig Beta 0.3 release [4], converted to one row per word following [28]. Each word carries its normalised form, its transliteration split into signs, its language layer and, where TLHdig provides a morphological analysis, a word class. The release is downloaded by script and checked by its MD5 sum, so the token table can be rebuilt from the public data alone. As in [5], reference manuscripts must have at least 17 legible words.

Orphans. Orphans are all CTH 470 manuscripts and all EBo manuscripts, wherever TLHdig files them (1,919 and 1,151 manuscripts; 3,033 in total). The experiments load the reference CTHs, CTH 470 and the two pools CTH 670 and CTH 832; these contain 2,899 orphans, of which 2,010 have at least five legible words and are placed (median 7 words over all loaded orphans). Orphans are never part of the reference.

Reference traditions. The abstract that announced this study proposed CTH 670, 701 and 832 as the reference. In TLHdig, however, CTH 670 and 832 are where the EBo fragments are filed, so they are pools of orphans rather than traditions; as a three-class problem they are also dominated by CTH 670 (majority baseline 78%). We therefore define the reference from the literature (Table 1). The Hurrian festival groups follow Table 3 of [5]. Its single Luwian group is split as the HFR edition modules split it, into the festivals of Ištanuwa and Lallupiya and those of Ḫuwaššanna, and HFR's Palaic festivals are added [12]. The ritual traditions are the Kizzuwatna rituals as delimited by Miller [6]; the western (Arzawa) rituals, that is, the augur rituals [9] and the rituals whose colophon names Arzawa or Ḫapalla as the practitioner's home; the Luwian rituals of Puriyanni, Kuwattalla and Šilalluḫi (CTH 758–763), which Yakubovich and Mouton place in the Lower Land rather than Kizzuwatna [32]; and the Hattian cult layer of Klinger [33], whose texts we take from the citations of his study in the Mainz Konkordanz [3]. The Ambazzi rituals (CTH 391, 429, 463) show Luwian influence but no stated link to Arzawa [8], so they are not part of the western group and are placed with the residue. The published grouping of [5] with two CTH-range controls (CTH 471–500 and CTH 390–469) is run as the baseline.

Two sets leave the reference and are placed like orphans. CTH 500, "fragments of festival and incantation rituals from Kizzuwatna", is a running list of fragments rather than a composition, and HFR counts it among its festival texts; and the texts of CTH 390–469 that no defining study claims form a residue. Both become test sets (Section 5.5).

CTH 701 labels. The subgroup labels of CTH 701 (Salvini and Wegner's groups a–f with their sub-numbered parallels [7]) and the script dates (Middle and New Script) come from the Mainz Konkordanz [3], parsed by script and frozen as a dated snapshot. Checked against the working sheet of [5], the groups agree for 120 of 121 shared fragments; the one difference is a missing group letter in the Konkordanz (KBo 64.168, listed as "701.$_2$.I"), corrected to b_2 in a documented table.

4. Method

Features. Two representations are built for every manuscript [38]. The word space holds tf-idf weights of normalised word forms, and represents what is written. The sign space holds tf-idf weights of sign n-grams ($n = 1$–3) taken within words, and represents how it is written. The combined space concatenates both. Weights are sublinear, rows are L2-normalised, and features occurring in fewer than two manuscripts are dropped.

Assignment. Each tradition is represented by the normalised centroid of its manuscripts, and a fragment is scored by its cosine similarity to every centroid. It is assigned to the most similar tradition when its lead over the runner-up (the *margin*) exceeds a threshold, and left unassigned otherwise.

Evaluation. Validation is leave-one-manuscript-out: each reference manuscript is removed from its own centroid, which can be done in closed form, and reassigned. We report accuracy and per-tradition recall against two baselines: always guessing the largest tradition (majority), and the same procedure with shuffled labels (200 permutations, which also give a p-value). The idf weights are fitted on the whole reference set (transductive idf).

Table 1
Reference traditions, the studies that define them, and the number of reference manuscripts with 17 or more legible words. CTH numbers are those of the Portal CTH [2].

Tradition	CTH	Defining studies	Criterion	*n*
Kizzuwatna–Lower Land (Hurrian)	628, 699–700, 711–713, 719, 722	Gordin et al. [5] Table 3; Wegner and Salvini [34]	Hurrian festivals whose cult geography is Kizzuwatna and the Lower Land, among them the *(ḫ)išuwa*-festival (CTH 628)	311
Syro-Mesopotamia (Hurrian)	698, 701, 714–715, 721	Gordin et al. [5] Table 3; Salvini and Wegner [7]	rituals of the AZU priest (CTH 701) and the Ištar/Šawuška cults of Nineveh (714–715) and the Amanus (721)	164
Mixed (Hurrian, Syrian, Luwian)	704–706, 720	Gordin et al. [5] Table 3	festivals with mixed Hurrian/Syrian/Luwian cult elements	198
Luwian festivals (Ištanuwa, Lallupiya)	665, 771–773	HFR-Team [12]; Görke [13]; Hutter [35]	HFR edition module "Luwian festivals": cult towns Ištanuwa and Lallupiya	50
Ḫuwaššanna festivals	690–694	HFR-Team [12]; Hutter [35]	cult of Ḫuwaššanna of Ḫubešna in the Lower Land (Laroche's titles); kept apart from Ištanuwa by HFR	180
Palaic festivals	750–754	HFR-Team [12]; Görke [13]	HFR edition module "Palaic festivals"	34
Kizzuwatna rituals	404, 471–499	Miller [6]; Strauß [36]; Görke [37]	practitioners from Kizzuwatna, Hurro-Luwian ritual language; Laroche's block plus Maštigga (CTH 404); the fragment collection CTH 500 excluded	164
Arzawa / western rituals	394, 402, 406, 410, 425, 433	Bawanypeck [9]; colophons	augur rituals (425, 433) and the rituals whose colophon names Arzawa or Ḫapalla as the practitioner's home (394, 402, 406, 410); the Ambazzi rituals (Christiansen 8) left the group, as they show Luwian influence but no stated link to Arzawa	11
Luwian rituals (Puriyanni, Kuwattalla)	758–763	Yakubovich and Mouton [32]	Luwian rituals of Puriyanni, Kuwattalla and Šilalluḫi, placed in the Lower Land, not Kizzuwatna	10
Hattian cult layer	591, 627, 725–729, 731–746, 820–821	Klinger [33]	texts cited by StBoT 37 in the Mainz Konkordanz: Monatsfest, KI.LAM, Hattian and Hattian–Hittite texts	478

Fragment size and abstention. To learn how small a fragment can be, manuscripts with at least 40 words (at most 40 per tradition) are held out in turn and cut into ten random windows of k words, $k \in \{3, 5, 8, 10, 17, 25, 40\}$. The abstention threshold is the smallest margin at which the held-out windows reach 80% precision; a bootstrap over the windows gives its 90% interval.

Layers and word classes. To separate cultural content from scribal habit, the word space is split into layers and each layer is evaluated alone: logograms (Sumerograms and Akkadograms), Hittite vocabulary, foreign-language words (Hurrian, Luwian, Hattian, Palaic), and the 382 words attested in more than one spelling, once with their spelling kept and once collapsed to the normalised form. If the spelling carried tradition beyond the word, the first would beat the second. Seven per-manuscript orthographic rates (shares of logograms, Akkadograms, phonetic complements, determinatives, plene

Table 2
Leave-one-manuscript-out accuracy of tradition assignment for three reference setups. Majority: always the largest class; shuffled: mean of 200 label permutations.

Reference setup	classes	n	majority	shuffled	words	signs	both
CTH 670 / 701 / 832 (abstract)	3	1,254	78%	55%	88%	89%	**89%**
StBoT 73 groups + CTH-range controls	6	1,637	23%	18%	69%	72%	**72%**
Literature-defined traditions (this paper)	10	1,600	30%	17%	81%	84%	**84%**

spellings, CVC signs and *ú*) are tested with a 10-fold linear discriminant analysis. Following the categories of Table 4 in [5] (agents, deities, function words, foreign words, realia and verbs), assigned from TLHdig's morphological analyses, we evaluate each word class alone and the text without it. Distinctive features per tradition are ranked by weighted log-odds with an informative Dirichlet prior [39], on manuscript frequencies.

Open set. The margin rule asks only which tradition is nearest. To measure how often it is fooled, we score texts whose genre is known and lies outside the reference, taken from Laroche's own genre blocks: myths, prayers and hymns, oracles, historical texts and treaties, instructions, cult inventories [40], festivals of traditions not in the reference, and the Luwian rituals of Taurisa (CTH 764–766), a further Luwian ritual tradition outside the reference. Reference and outside texts are cut into windows and scored in five folds. We measure the share of outside windows that pass the margin, how well the top similarity separates reference from outside windows (AUROC), and a *genre gate* in which the outside genres become classes of their own.

QTM coverage. For each tradition, the *archetype* is approximated by the 200 highest-weighted terms of its centroid in the combined space, and its *coverage* is the share of those terms attested in at least one orphan assigned to it (Section 6).

5. Results

5.1. Validation: do the traditions hold together?

Table 2 compares the three reference setups. The abstract's CTH 670/701/832 setup reaches 89%, but its majority baseline is already 78%, so it shows little. The published groups with CTH-range controls reach 72% over six classes. The literature-defined traditions reach 84% over 10 classes, against 30% for the majority baseline and 17% for shuffled labels ($p = 0.005$). Words and signs perform alike (81% and 84%).

Per-tradition recall (Table 3) ranges from 40% to 95%. The HFR split holds: the Ḫuwaššanna festivals (91%) and those of Ištanuwa and Lallupiya (84%) are kept apart by the texts themselves, so the single Luwian group of [5] joined two cult traditions. The weakest tradition is also the smallest: Luwian rituals (Puriyanni, Kuwattalla) has 10 manuscripts long enough to test. Removing all logograms lowers accuracy only from 81% to 75% (words) and from 84% to 81% (signs), so the signal does not rest on the logographic cult vocabulary alone.

5.2. How small can an orphan be?

Accuracy rises steeply with fragment size and then flattens (Figure 1). A three-word window is assigned correctly in 53% of cases, a ten-word window in 68%, and a forty-word window in 79%, against a chance level of 10%. Sign n-grams are slightly better than words on short windows (67% and 64% at ten words). The abstention margin calibrated on these windows is 0.016 (90% interval 0.015–0.017); above it, 80% of windows are correct, and 69% of them pass.

Table 3
Per tradition: reference manuscripts, leave-one-out recall in each space, orphans assigned (Section 5.5), and QTM coverage of the 200 core terms by those orphans (Section 6).

Tradition	*n*	words	signs	both	orphans	coverage
Kizzuwatna–Lower Land (Hurrian)	311	80%	82%	83%	77	93%
Syro-Mesopotamia (Hurrian)	164	72%	66%	67%	62	92%
Mixed (Hurrian, Syrian, Luwian)	198	61%	67%	67%	25	66%
Luwian festivals (Ištanuwa, Lallupiya)	50	84%	82%	84%	8	57%
Ḫuwaššanna festivals	180	88%	91%	91%	14	56%
Palaic festivals	34	79%	82%	82%	4	34%
Kizzuwatna rituals	164	87%	91%	92%	574	100%
Arzawa / western rituals	11	73%	64%	64%	12	54%
Luwian rituals (Puriyanni, Kuwattalla)	10	50%	40%	40%	22	67%
Hattian cult layer	478	90%	94%	95%	142	98%

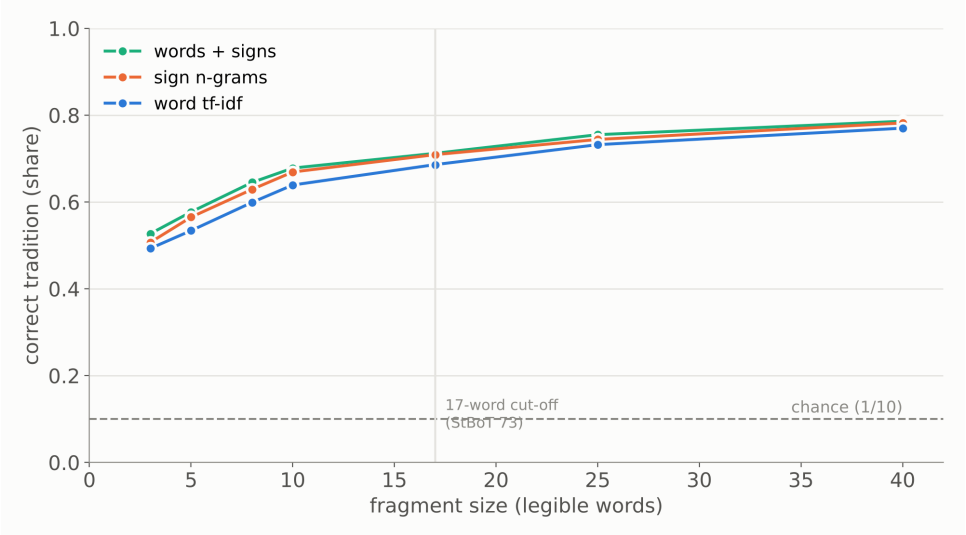


Figure 1: Accuracy of tradition assignment for held-out windows of k legible words. The vertical line marks the 17-word cut-off of [5].

Table 4
Each layer of a text alone: leave-one-manuscript-out accuracy over the reference traditions. *n*: manuscripts with at least one feature of the layer.

Layer	*n*	majority	shuffled	accuracy
Logograms (Sumero- and Akkadograms)	1,535	28%	13%	**70%**
Hittite vocabulary	1,530	27%	15%	**71%**
Foreign-language words (Hurrian, Luwian, Hattian, Palaic)	396	50%	24%	**88%**
Variable words, spelling collapsed	1,509	28%	11%	**60%**
Variable words, spelling kept	1,505	28%	13%	**61%**
Seven orthographic rates (LDA, 10-fold)	1,601	30%	30%	**39%**

5.3. What identifies a tradition: what is written, or how?

Table 4 evaluates each layer alone. Content carries the identity of a tradition: logograms reach 70%, Hittite vocabulary 71%, and foreign-language words 88% in the 396 manuscripts that have them. The variable words reach 61% with their spelling kept and 60% with it collapsed: the spelling choice adds nothing beyond the words themselves. The seven orthographic rates are a weak but real signal, 39% against 30% for shuffled labels ($p = 0.0099$).

Table 5
Word classes: accuracy with only that class, and with everything except it, in the word and sign spaces. Majority baseline for the class alone.

			class alone		without class	
Class	*n*	majority	words	signs	words	signs
Agents	1,115	31%	52%	60%	80%	83%
Deities	1,141	24%	59%	59%	79%	82%
Function words	1,499	27%	56%	57%	81%	84%
Verbs	1,454	27%	55%	53%	80%	84%
Realia	1,509	27%	71%	73%	76%	79%
Foreign-language words	405	49%	87%	89%	78%	80%

The word classes of [5] point the same way (Table 5). Removing function words, verbs or agents costs almost nothing (81% with all words, 81% without function words, 80% without verbs): these classes form a layer shared across traditions. Alone they still reach 56% and 55%, well above the majority baseline, so the formulae differ between traditions, but less than realia (71%) or foreign words (87%). The previous study suggested that shared Akkadograms among function words and verbs might reveal a writing habit [5]; at the scale of traditions, that layer is common ground rather than a marker.

Cultural and scribal profiles. The distinctive features of each tradition (Appendix A) are ritual materials, offerings, performers and deities: AZU, EREN and SÍSKUR in the Syro-Mesopotamian group, SÍG and ZA.GÌN in the Kizzuwatna rituals, LUGAL and DUMU.É.GAL in the Hattian cult layer. Spelling habits behave differently. The plene spelling *me-ma-a-i* appears in 18 of 46 Kizzuwatna–Lower Land manuscripts, all of them CTH 628, 712, and *pé-e-ra-an* in 5 of 19 Ištanuwa and Lallupiya manuscripts, all CTH 665. Spelling habits attach to compositions and their copies, not to whole traditions. Inside CTH 701, the orthographic rates do not predict script date either (75% against a majority of 83%), so there they are not merely a proxy for date. The per-tradition rates are in Appendix B.

5.4. CTH 701 as a test bed

Salvini and Wegner's subgroups of the AZU rituals provide partial ground truth. On 82 manuscripts in 8 subgroups, the method recovers 58% of labels (words 50%, signs 60%) against a majority baseline of 21% ($p = 0.005$; Figure 2). The same procedure does not predict script date: 74% against a majority of 83% ($p = 0.4527$). The subgroups are therefore not script-date clusters. Removing the Hurrian and other foreign words lowers subgroup accuracy from 50% to 47%, the largest drop of any class, yet the remainder is still more than twice the majority baseline: the recitations carry the subgroups, but the Hittite frame text differs between them as well. The 50 unclassified and group-f fragments are placed among groups a–e as candidate attributions in the released tables.

5.5. Orphans placed

With the calibrated margin, 940 of the 2,010 orphans are assigned and 1,070 are left unassigned (889–981 assigned across the margin's 90% interval; Figure 3). The Kizzuwatna rituals receive 574 of them (61%).

The two test sets show why that number needs care. The CTH 390–469 residue is placed mostly with the Kizzuwatna rituals (152 of 186 assigned), although no study assigns these texts there: the Kizzuwatna rituals are the only large ritual class, so the placement follows genre before origin. CTH 500 behaves differently. Only 54 of its 128 assigned fragments go to the Kizzuwatna rituals and 73 to the Hurrian festival groups, which agrees with HFR's treatment of CTH 500 as festival material [12].

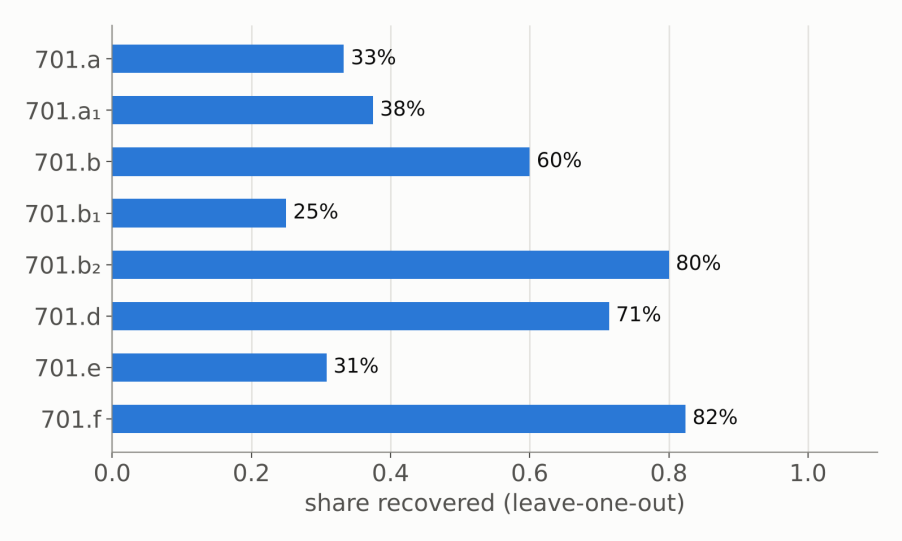


Figure 2: Recall of Salvini and Wegner's CTH 701 subgroups, leave one manuscript out.

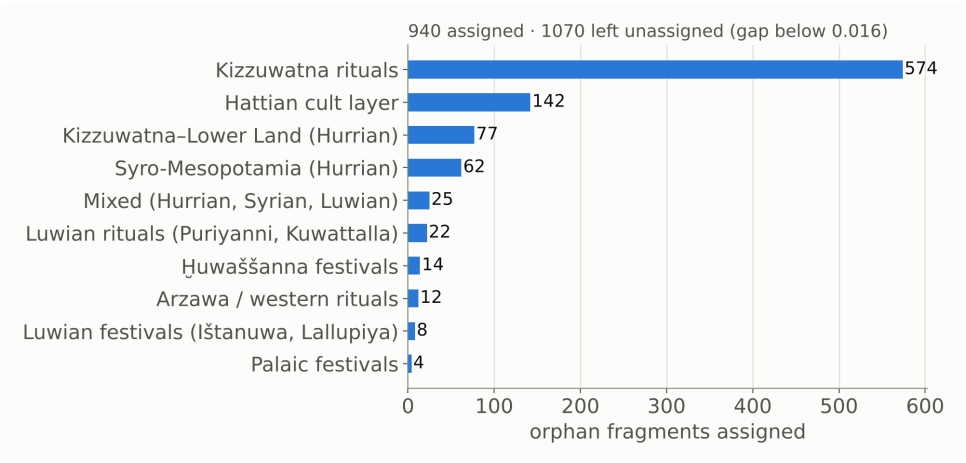


Figure 3: Orphans (CTH 470 and EBo, five or more legible words) assigned to each tradition above the calibrated margin.

5.6. Does a fragment fit any known tradition?

Table 6 shows that the margin rule is fooled by texts from outside the reference. At 10 words, it still places 28%–37% of the windows from the six other genres in some tradition, and 64% of the windows from festivals of other traditions. A floor on similarity separates other genres fairly well (AUROC 0.77–0.88) but not other festivals (0.56). A genre gate works better: with the outside genres as classes of their own, 86%–96% of their windows and 69% of the other festivals are recognised as outside, at the cost of sending 19% of reference windows away. The Luwian rituals of Taurisa are placed in 21% of windows and recognised by the gate in 98%, but rest on only 14 texts.

Applied to the orphans, the gate finds 255 of 2,010 that are closer to an outside genre than to any tradition by more than the margin; 96 of them had been assigned. Most are Hattian-layer placements that look like festivals of other traditions (29) and Kizzuwatna placements that look like myths (14) or instructions (18). They are released as candidates for re-filing.

Table 6
Open-set test, windows of 10 words, five folds. Placed: share passing the margin rule (wrongly, for outside texts). AUROC: separation of reference and outside windows by top similarity. Gate: share whose nearest class is an outside genre.

Texts	*n*	placed	AUROC	gate
Myths	300	31%	0.81	86%
Historical texts, treaties	300	33%	0.83	96%
Cult inventories	300	28%	0.86	90%
Festivals of other traditions	300	64%	0.56	69%
Oracles	300	30%	0.88	96%
Prayers and hymns	159	28%	0.87	95%
Instructions	112	37%	0.77	92%
Luwian rituals of Taurisa	14	21%	0.92	98%
Reference traditions (held out)	1,601	73%	–	19%

6. Quantitative Scribal Tradition Mapping

The results suggest a division of labour between two levels of a text, which QTM makes explicit. Borrowing the terms of textual criticism [41, 42], a tradition is treated as an *archetype*: not a reconstructed text, but the body of vocabulary, formulae and foreign-language recitation that its manuscripts share. Each manuscript, and each orphan, is a *witness* that attests a part of it. What identifies the archetype is its content: the layers that carry tradition in Section 5.3 are the cultural ones. What marks the witness is its spelling: plene habits and orthographic rates cluster by composition and copying line, not by tradition, which agrees with the view that Hittite writing habits belong to scribal circles and their schooling [16]. Logograms sit on both sides: the object named is cultural, the decision to write it logographically is scribal [17].

An orphan is a low-coverage witness. Coverage, the share of a tradition's 200 core terms that its assigned orphans attest, is near complete for the traditions that receive many orphans (100% for the Kizzuwatna rituals, 98% for the Hattian layer) and low for those that receive few (34% for the Palaic festivals with 4 orphans). In its present form this measure grows with the number of orphans, so it cannot yet be compared across traditions; rarefaction to equal orphan counts [43], and unseen-species estimates of the kind applied to lost medieval literature [44], are the natural next steps. Read this way, the question becomes historical: which traditions circulated widely in copies that now survive only as fragments, and which are known almost only from their identified compositions.

7. Discussion and limitations

Genre before origin. The residue result shows the main risk of the method. A fragment is placed by what it shares with the reference, and the largest shared layer is genre. A Hittite ritual from outside every reference tradition will look most like the largest ritual class. The open-set test shows the same risk across genres. A placement should therefore be read as "nearest of these" unless the reference includes the plausible alternatives as classes. The genre gate is a first step, and each new tradition narrows the gap: with the Luwian rituals of CTH 758–763 in the reference, 13 residue texts are placed with them rather than with a larger ritual class.

Limitations. TLHdig's normalised forms are uneven across editors, and broken words are dropped, so short fragments over-represent formulaic lines. The idf weights are fitted on the whole reference set. Several traditions are small (Luwian rituals (Puriyanni, Kuwattalla) has 10 test manuscripts), and their recall is uncertain. Festival manuscripts differ by function (outline tablets, day tablets, cult inventories close to festival text [40]), which word profiles may track alongside tradition. The Hattian layer's preference for plene writing and few logograms may reflect Old Hittite manuscripts as much as tradition,

and is still to be checked against script dates. Finally, every placement is a hypothesis for philological verification on the tablet: an orphan placed near a known manuscript is a candidate join, not a join.

8. Conclusion

Content-based assignment places Hittite ritual fragments in literature-defined traditions well above chance, from about ten legible words on, and recovers subgroup structure inside CTH 701 without being a proxy for script date. The signal lies in what is written; spelling marks the copying line. A closed reference over-assigns, and the open-set test shows how to make the abstention honest. QTM turns these observations into a programme: orphans as witnesses, traditions as archetypes, and coverage as the measure of how much of a tradition survives only in fragments.

9. Data and code availability

Code, the frozen CTH 701 label snapshot and all outputs are in the repository https://github.com/shaigordin/measuring-scribal-literacy (folder `orphans/`). `reproduce.sh` downloads TLHdig 0.3 from Zenodo, verifies its checksum and rebuilds every experiment, figure and table; all randomness is seeded and a rerun is byte-identical. The paper itself (`paper/`) is generated from those outputs: `make_numbers.py` writes every number in the text and every results table, and `generated/numbers_index.csv` lists each number with the file and field it comes from.

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Declaration on Generative AI

During the preparation of this work, the author used Claude (Anthropic) in order to: write and run the analysis code, draft the text, and check bibliographic references. After using this tool, the author reviewed and edited the code and the content as needed and takes full responsibility for the publication's content.

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A. Distinctive features per tradition

B. Orthographic profiles

Table 7
The four most distinctive features of each tradition per layer (weighted log-odds with an informative prior [39], on manuscript frequencies), as TLHdig forms. Full lists in `E7_distinctive_features.csv`.

Tradition	Logograms	Hittite vocabulary	Foreign words
Kizzuwatna–Lower Land (Hurrian)	EGIR-ŠÚma, EGIR-ŠUma, UP-NI, BA.BA.ZA	½, manuzia, nupatik, šipantanzi	tai, anaḫi, tea, manuzunna
Syro-Mesopotamia (Hurrian)	AZU, EREN, SÍSKUR, AD.KID	ḫurlili, ḫuprušḫia, aḫrušḫiaz, aḫrušḫia	kelum, kelu, DINGIRna, KI.MIN
Mixed (Hurrian, Syrian, Luwian)	KI.MIN, DINGIRna, NINDA.SIG, EGIRantama	ḫepat, ekuzi, ninatta, kulitta	DINGIRna, allai, ~er, šiššie
Luwian festivals (Ištanuwa, Lallupiya)	GAL-ŠU-NU, ZA.LAM.GAR, KA.GAG, DUMU.LUGAL	ḫuḫupal, kimri, anianzi, 9	wašu, anta, zanta, paiu
Ḫuwaššanna festivals	BE-EL, KAŠ, NAR, SÌR-RU	paluaizi, akuanna, takarmun, pianzi	arešti, wa~, taḫarua, taḫapuazi
Palaic festivals	GEŠTINit, UTUi, A-NA NINDA.GUR$_4$.RA, ŠU.GIma	išnuri, ziparua, kataḫzipuri, memaḫḫi	taparnaš, ânta, ni, kuiša
Kizzuwatna rituals	SÍG, ZA.GÌN, ŠU.GI, SA$_5$	tarpalaš, ḫuittianzi, našma, kureššar	zuli, enna, ari, DINGIRnaša
Arzawa / western rituals	A-NA KAL, MUŠEN.DÙ, KAL, TU$_7$	kuršaš, paimi, eš, ~ḫi	–
Luwian rituals (Puriyanni, Kuwattalla)	EMEan, SAG.DUi, MUan, ŠU.GIma	pankauaš, karpin, ~ša, ~tat	–
Hattian cult layer	LUGALi, LUGALuš, DUMU.É.GAL, GIŠ.INANNA	ta, ḫalzai, wezi, petai	taparna, pala, katte, ma

Table 8
Mean orthographic rates per tradition as *z*-scores over all reference manuscripts: shares of logograms, Akkadograms, phonetic complements, determinatives, plene spellings, CVC signs and *ú*.

Tradition	logo.	Akkad.	compl.	det.	plene	CVC	*ú*
Kizzuwatna–Lower Land (Hurrian)	+0.23	+0.42	-0.20	+0.04	-0.33	+0.03	-0.28
Syro-Mesopotamia (Hurrian)	+0.01	+0.12	-0.27	-0.04	+0.30	-0.20	-0.32
Mixed (Hurrian, Syrian, Luwian)	+0.30	-0.24	+0.19	+0.31	-0.21	+0.27	-0.11
Luwian festivals (Ištanuwa, Lallupiya)	-0.18	+0.35	-0.07	+0.11	-0.28	+0.01	+0.25
Ḫuwaššanna festivals	+0.33	+0.28	-0.25	+0.38	-0.34	+0.07	-0.15
Palaic festivals	-0.74	-0.20	+0.72	-0.07	+0.11	-0.05	+0.11
Kizzuwatna rituals	+0.13	+0.07	-0.21	-0.45	-0.29	+0.16	+0.09
Arzawa / western rituals	+0.07	+0.25	+0.15	-0.00	-0.34	+0.44	+0.06
Luwian rituals (Puriyanni, Kuwattalla)	-0.13	-0.07	+1.40	-0.58	-0.82	+0.42	+0.10
Hattian cult layer	-0.37	-0.37	+0.23	-0.13	+0.47	-0.16	+0.33